

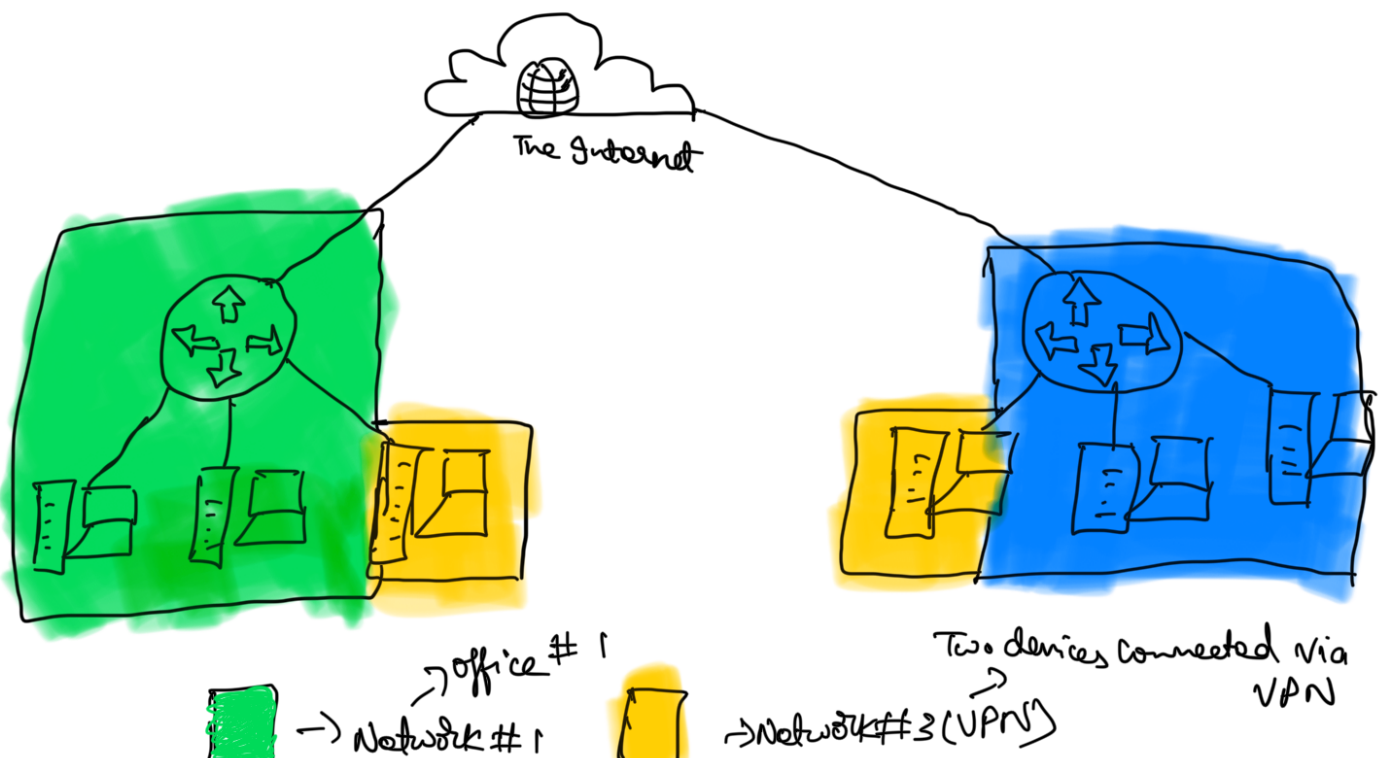
VPN Basics :-

A Virtual private Network (VPN for short) is a technology that allows devices on separate networks to communicate securely by creating a dedicated path b/w each other over Internet. Devices connected within this tunnel form their own private network.

* Instead of being restricted to communicating only within your local office network, a VPN allows two completely different office networks to connect and talk to each other as if they were on same private network.

Note :- is a cybersecurity tool that creates an encrypted, private tunnel b/w your device and Internet. It protects your data from hackers, hides your real IP address and masks your physical location.

Ex :- your device within same network such as within a business can directly communicate. However, a VPN allows two offices to be connected. Let's take diagram below where there are 3 networks.



→ Network #2 office #2

The devices connected on network #2 are still part of network #1 and network #2 but also join together to create a private network (Network #3) that only devices that are connected via VPN can communicate over.

Benefits of VPN :-

Description

1) Allows networks in different geographical locations to be connected	For example, a business with multiple offices will find VPN's beneficial, as it means that resources like servers / infrastructure can be accessed from another office.
2) offers privacy	VPN technologies use encryption to protect data. This means only intended devices it was being sent from and is destined for, meaning data isn't vulnerable to sniffing.
3) offers Anonymity	Journalists and activists depends upon VPN's to safely report on global issues in countries where freedom of speech is controlled. Usually, your traffic can be viewed by your ISP and other intermediaries and therefore tracked. The level of anonymity a VPN provides is only as much as how other devices on network respects privacy. Ex :- VPN that logs all of your data / history is essentially the same as not using VPN in this regard.

Try hack me uses VPN to connect you to our vulnerable machines without making directly available on Internet! This means that:

- you can securely interact with our machines.
- Service providers such as ISP don't think you are attacking another machine on internet.
- VPN provides security to try hack me as vulnerable machines are not available using Internet.

VPN technologies improved over years :- below

VPN technology	Description
	This technology is used by PPTP to allow for

PPP	Authentication and provide encryption of data. VPN work by using private key and public certificate (similar to SSH). A private key and certificate must match for you to connect. This technology is non scalable.
PPTP	Point to point tunneling protocol (PPTP) is technology that allows data from PPP to travel and leave a network. PPTP is very easy to setup and supported by most devices. However weakly encrypted in comparison to alternatives.
IPSec	Internet protocol security (IPSec) encrypts data using existing Internet protocol (IP) framework. IPSec is difficult to set up in comparison to alternatives however it's successful, it boasts strong encryption and is also supported on many devices.

Quiz:-

1) What VPN technology only encrypts & provides authentication of data?

A) PPP

2) What VPN technology uses IP framework?

A) IPSec